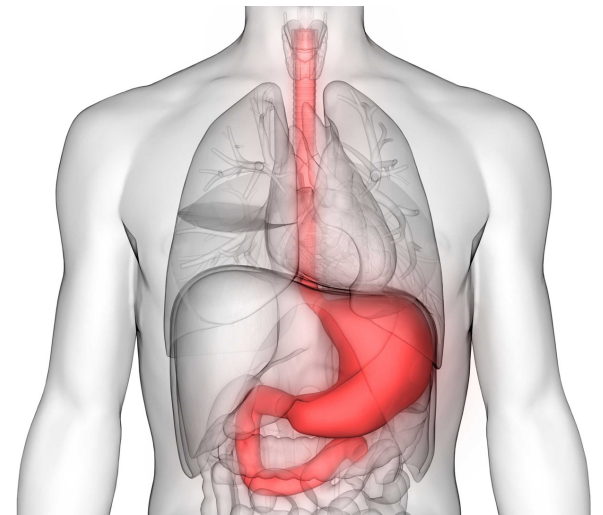
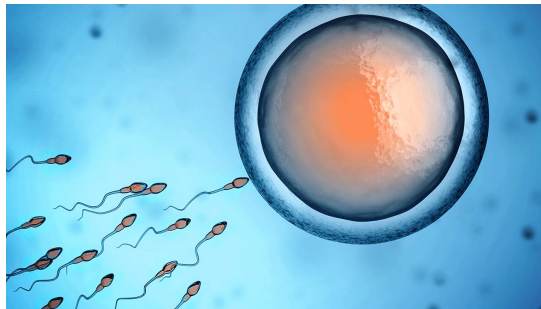
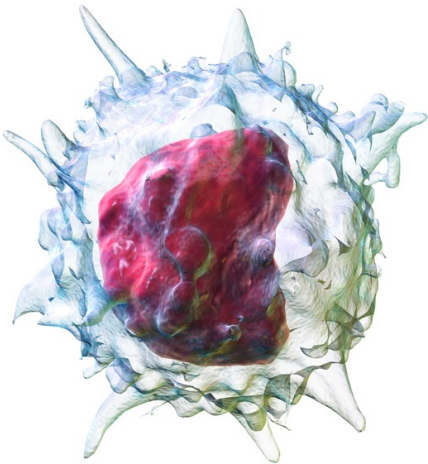
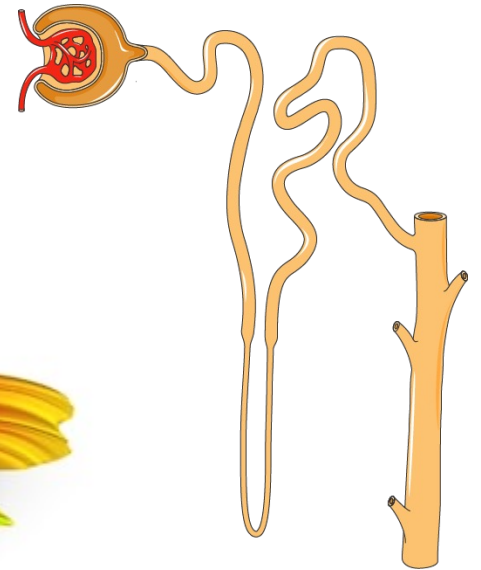
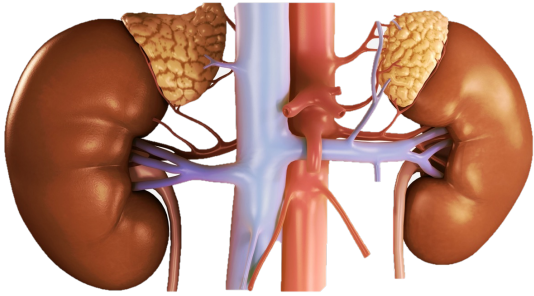
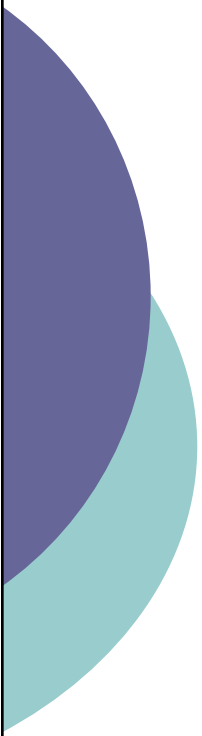


Welcome to Bio 40C!

Jeff Schinske



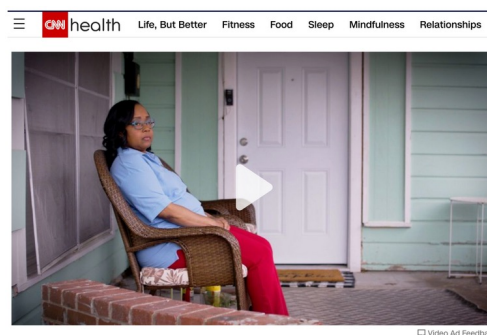


Bio 40C Proposed Norms

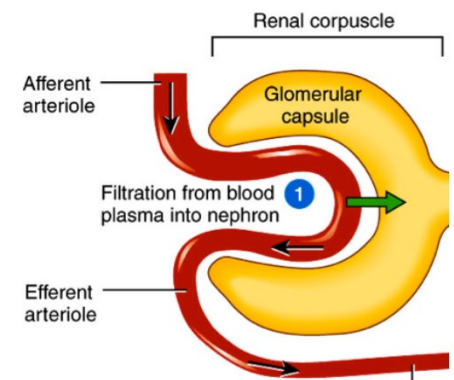
- Seek to enjoy your colleagues and your experiences in class!
- Actively select language that is inclusive, respectful, and welcoming
- Assume good intentions...no heroes and no villains (in this class)
- Recognize that we all have strengths as well as areas for growth
- Be curious, be skeptical
- Ask for help (and expect to receive it!)

Share with a neighbor...

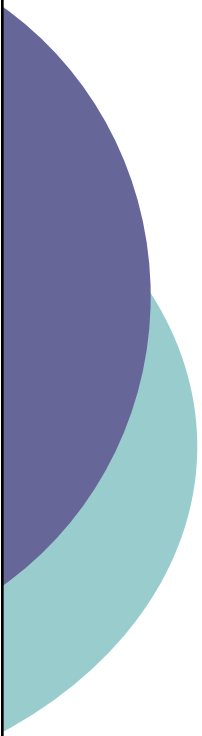
1. Name
2. Something important to know about who you are
3. What most stood out to you from the last week of Canvas activities?



Can a formula be racist? She says one put her health at risk



Person with 1st name closest to start of alphabet goes first



Debriefing Last Week

○ Last Week's Canvas Work

- "I enjoyed the scientist spotlight and learning about eGFR. What really stood out to me this week was discussing the disparities among race in the health field."
- "I look forward to learning more about the filtration system because as of yet it is still quite mysterious to me albeit very fascinating considering beforehand I had no idea what happened to the water we drink."
- "I had a pretty rough week this week. Just feeling under the weather made my head a bit foggy."
- "I hope to have my work and study life balance in order with me starting a new job."
- "This week was actually harder for me compared to the first week because it feels like we are getting into harder topics with more dense information. The application questions are also getting harder to answer but I think it is cool that the application questions are applicable to our everyday life."

Antibiotics & the Digestive System

Antibiotics are used to treat bacterial infections. Some antibiotics target only very specific subspecies of bacteria, while others kill a wide range of bacteria, including bacteria in our intestines. One of the main side effects of broad spectrum antibiotics is diarrhea (watery bowel movements).

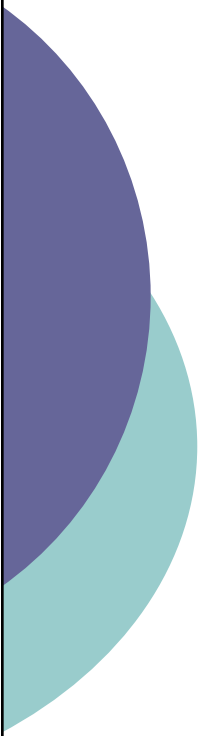
Think about the functions of the large intestine and the digestive activities of the bacteria that live there. Explain how diarrhea could occur as a result of killing off a large amount of your large intestine bacteria with antibiotics.

How could we make this more scientifically accurate?

What is the bacteria's function? How related to osmosis?

"Antibiotics can kill off good bacteria in addition to harmful bacteria. If those good bacteria cannot carry out their function in the colon, then there will be an imbalance that will disrupt osmosis and cause watery stool."





A Brief Preview of This Week

- Today

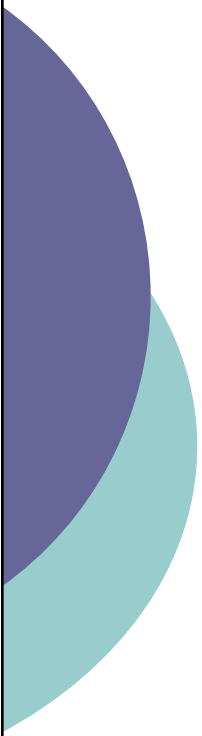
- Extending our knowledge of kidney filtration and its clinical importance

- Lab

- Urinary Anatomy

- Online Lecture Work This Week

- Exploring the way our kidneys make concentrated urine and help us avoid dehydration
- Exercising quantitative thinking in viewing a historical kidney study



What will you be able to do?

After class, you will be able to:

- Describe the impacts of blood pressure on glomerular filtration and kidney function.
- Justify the use of ACE inhibitors as medications for high blood pressure.

Kidney (glomerular) filtration is a key factor in human health



And our means of measuring filtration are intertwined with issues of racial justice.

What impacts glomerular filtration and what can cause kidney failure?

High Blood Pressure (Hypertension) and Kidney Failure

The Mercury News 

THINGS TO DO > ENTERTAINMENT • NewsMay 2023

Tina Turner admitted she put her life in danger by neglecting kidney disease, hypertension

Two months before her death, the legendary singer urged people to take high blood pressure seriously, explaining how it seriously damaged her kidneys to the point she needed a transplant

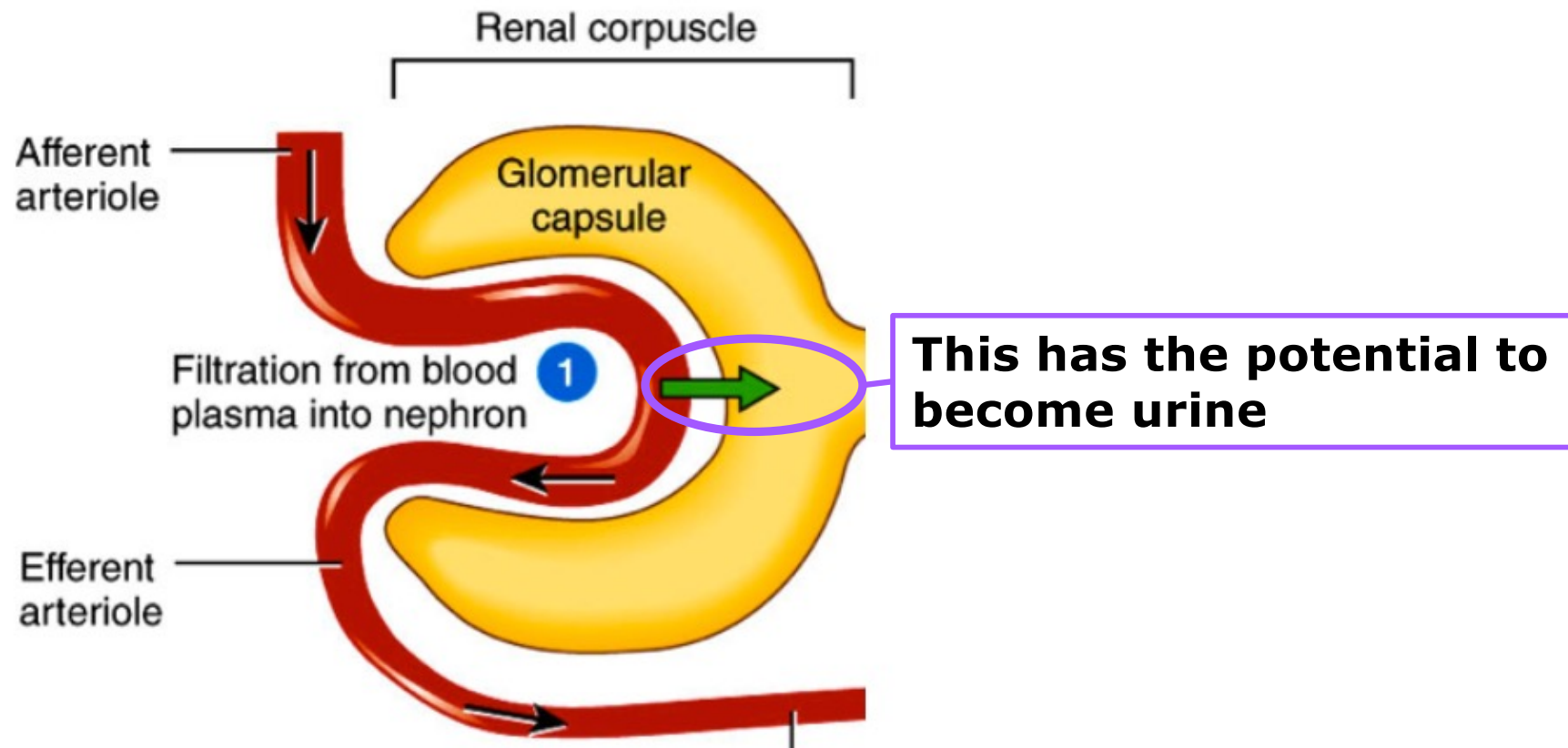


FILE – Tina Turner performs in a concert in Cologne, Germany on Jan. 14, 2009. Turner, the unstoppable singer and stage performer, died Tuesday, after a long illness at her home in Küsnacht near Zurich, Switzerland, according to her manager. She was 83. (AP Photo/Hermann J. Knippertz, file)

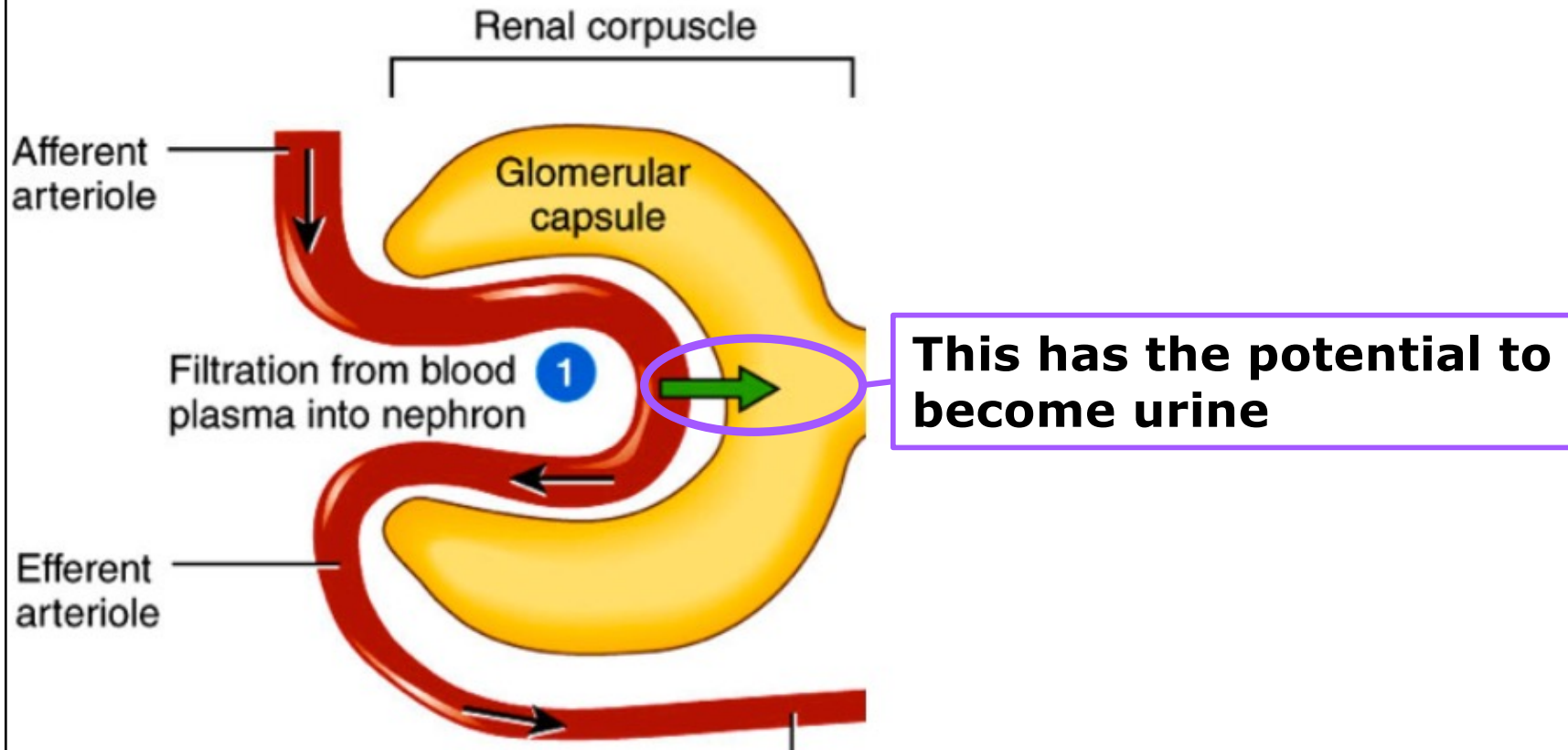
How does high blood pressure impact kidney function?

In this context, why is it so important for us to be able to accurately measure GFR?

You have hundreds of thousands of these in each kidney!



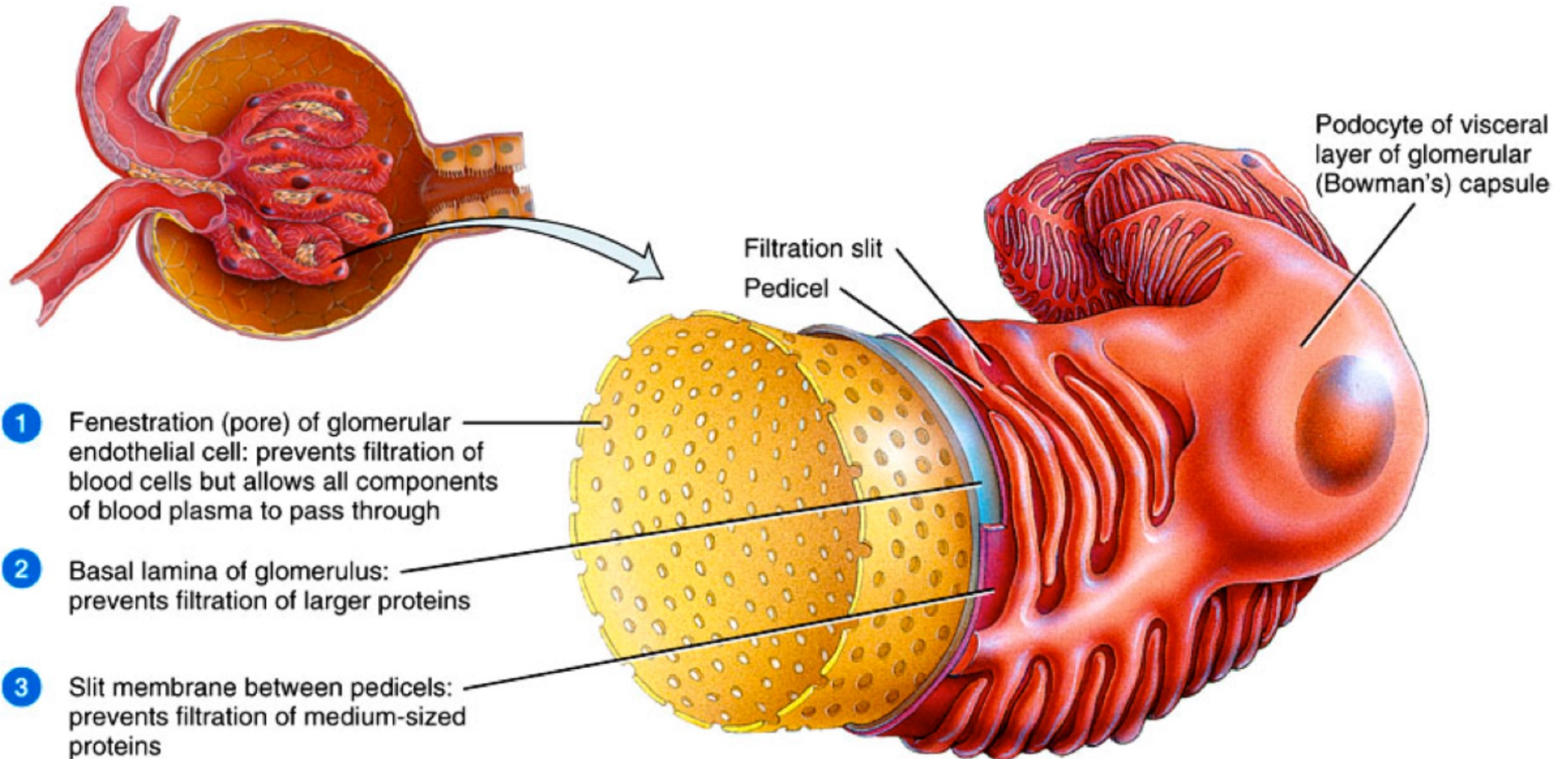
We're going to make some models of this!



If blowing into the model is difficult for you health-wise, don't do that part.

Please be careful where your model is pointed and be hygienic in terms of saliva!

What gets through the filter?



High Blood Pressure (Hypertension) and Kidney Failure

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How does high blood pressure impact kidney function?

In this context, why is it so important for us to be able to accurately measure GFR?

Use the following terms in your response:

Afferent arteriole
Proteins
Kidney Failure

5 **Your Name**

Your responses...

Doctors often treat chronic high blood pressure by prescribing drugs called Angiotensin Converting Enzyme (ACE) Inhibitors. These drugs help lower blood pressure and maintain proper kidney function.

Two white plastic bottles of oral tablets are shown side-by-side. The bottle on the left is labeled 'Fosinopril Sodium Tablets, 40 mg' and '90 Tablets' by Sandoz. The bottle on the right is labeled 'Captopril Tablets, USP' and '25 mg' by West-Wall Pharmaceutical Inc.

Hypertension, Kidneys, & ACE Inhibitors...

Renin-Angiotensin-Aldosterone System

